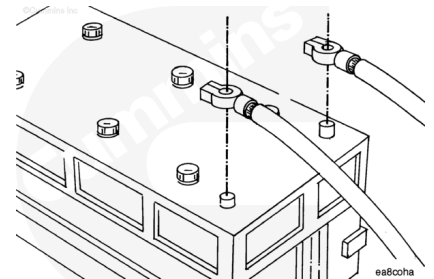


Trainings ▾

Resistance Check

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



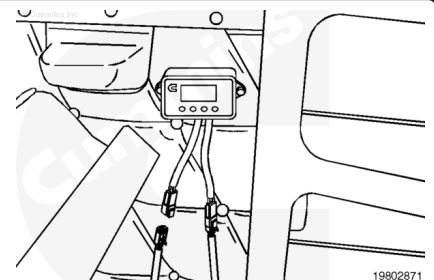
Turn the keyswitch to the OFF position. Disconnect the vehicle battery cables from the battery.

Note : Disconnecting the vehicle battery connections can require that the engine ECM real-time clock be reset using INSITE™.

Disconnect the cab thermostat jumper harness from the cab thermostat.

Disconnect the ICON™ idle control module A connector.

Set the multimeter to measure resistance.



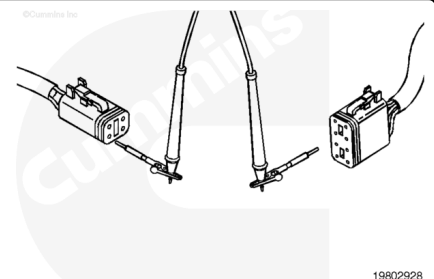
⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.

Touch one of the multimeter leads to pin 3 of the cab thermostat jumper harness connector (on the end that connects to the thermostat).

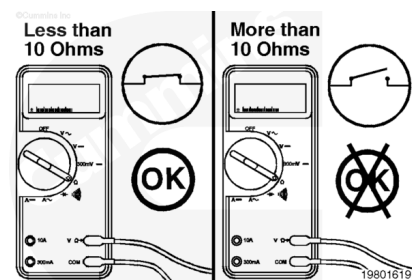
Touch the other multimeter lead to pin 4 of the ICON™ idle control module A harness connector.

Read the value displayed on the multimeter display.



The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If the circuit is **not** closed, isolate the problem area to the cab harness or ICON™ engine harness portion of the circuit.

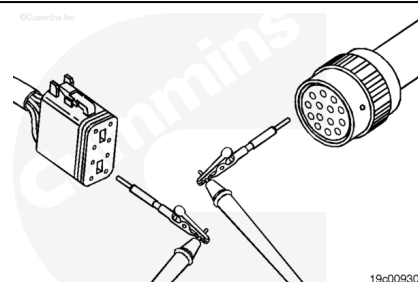


Disconnect the 14-pin pass-through connector at the vehicle's firewall.

Touch one of the multimeter leads to pin 4 of the ICON™ idle control module A harness connector.

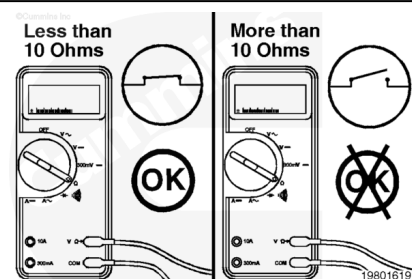
Touch the other multimeter lead to pin B of the 14-pin pass-through connector, engine harness side.

Read the value displayed on the multimeter display.



The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit. If the circuit is **not** closed, repair or replace the ICON™ engine harness. Refer to Procedure 019-200 (/qs3/pubsys2/xml/en/procedures/97/97-019-200.html), 019-208 (/qs3/pubsys2/xml/en/procedures/97/97-019-208.html), or 019-043. (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html)

If the circuit is closed, isolate the problem to the cab harness or cab thermostat jumper harness.

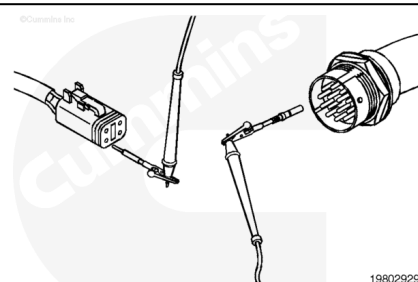


Disconnect the cab thermostat jumper harness from the cab harness.

Touch one of the multimeter leads to pin 3 of the cab harness thermostat harness connector, the 4-pin connector that is attached to the cab harness.

Touch the other multimeter lead to pin B of the 14-pin pass-through connector, cab harness side.

Read the value displayed on the multimeter display.



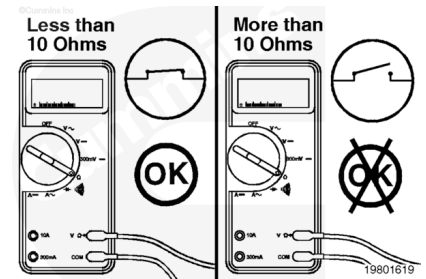
The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If the circuit is **not** closed, repair or replace the cab harness.

Refer to Procedure 019-200

(/qs3/pubsys2/xml/en/procedures/97/97-019-200.html), 019-

204 (/qs3/pubsys2/xml/en/procedures/97/97-019-204.html), or 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).

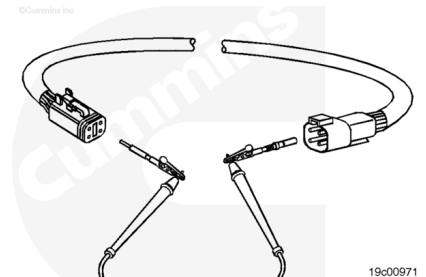


Check the cab thermostat jumper harness.

Touch one of the multimeter leads to pin 3 of the cab thermostat jumper harness connector on the end that connects to the thermostat.

Touch the other multimeter lead to pin 3 of the cab thermostat jumper harness connector on the end that connects to the cab harness.

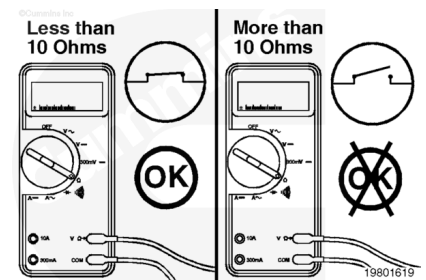
Read the value displayed on the multimeter.



The multimeter **must** display a reading of less than 10 ohms, which is a closed circuit.

If the circuit is **not** closed, repair or replace the cab thermostat jumper harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/97/97-019-204.html) or 019-295 (/qs3/pubsys2/xml/en/procedures/97/97-019-295.html).

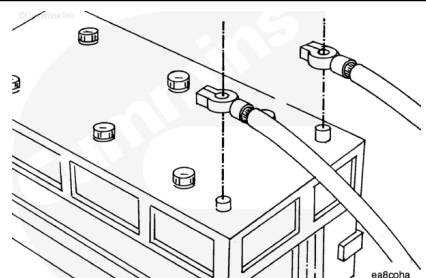
Connect all components after completing the repair.



Check for Short Circuit to Ground

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Turn the keyswitch to the OFF position. Disconnect the vehicle battery cables from the battery.

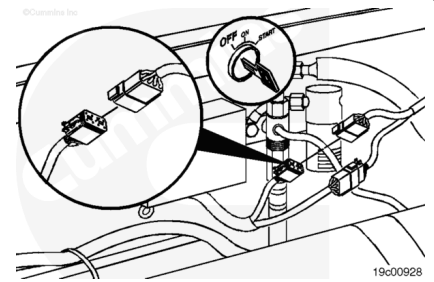
Note : Disconnecting the vehicle battery connections can require that the engine ECM real-time clock be

reset using INSITE™.

Disconnect the ICON™ idle control module A connector.

Disconnect the cab thermostat jumper harness from the thermostat.

Set the multimeter to measure resistance.



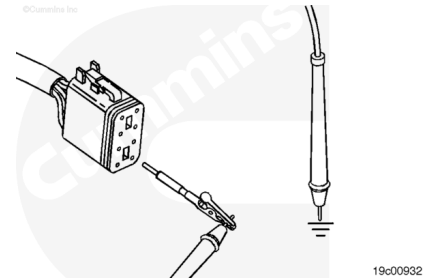
⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.

Touch one of the multimeter leads to pin 4 of the ICON™ idle control module A harness connector.

Touch the other multimeter lead to ground.

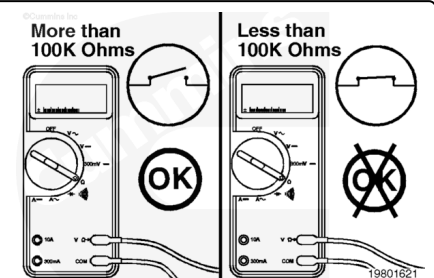
Read the value displayed on the multimeter display.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit to ground.

Isolate the short circuit to the cab harness, cab thermostat jumper harness, or ICON™ engine harness portion of the circuit.

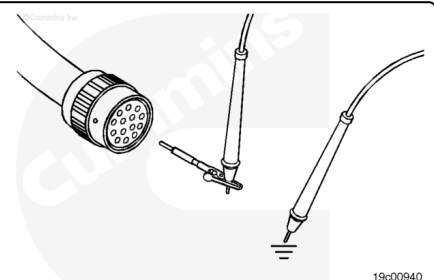


Disconnect the 14-pin pass-through connector at the vehicle's firewall.

Touch one of the multimeter leads to pin B of the 14-pin pass-through connector, engine harness side.

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter display.

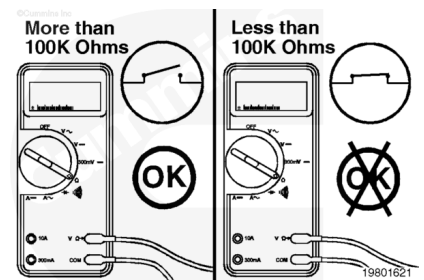


The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit to ground in the ICON™ engine harness. Repair or replace the ICON™ engine harness. Refer to Procedure 019-200

(/qs3/pubsys2/xml/en/procedures/97/97-019-200.html) or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

If the circuit is open, the short circuit is on the cab harness/cab thermostat harness side of the circuit. Isolate the short to the appropriate harness.

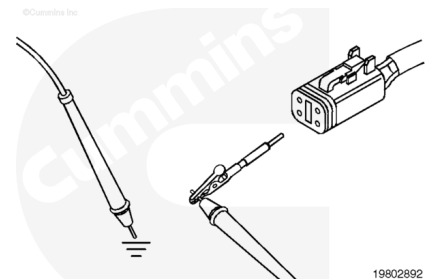


Disconnect the cab thermostat jumper harness connector from the cab harness.

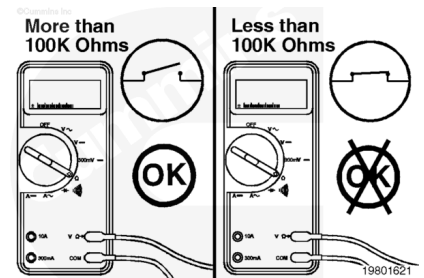
Touch one of the multimeter leads to pin 3 of the cab thermostat harness connector (on the cab harness).

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit. If the circuit is **not** open, there is a short circuit to ground in the cab harness. Repair or replace the cab harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/97/97-019-204.html) or 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).



If the circuit is open, the short is in the cab thermostat jumper harness. Repair or replace the cab thermostat jumper harness. Refer to Procedure 019-204

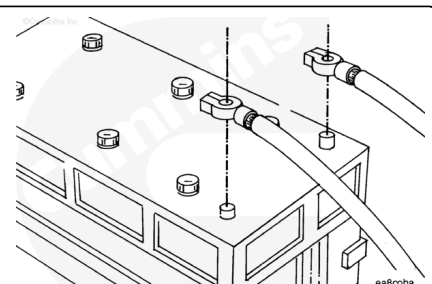
(/qs3/pubsys2/xml/en/procedures/97/97-019-204.html) or 019-295 (/qs3/pubsys2/xml/en/procedures/97/97-019-295.html).

Connect all components after completing the repair.

Check for Short Circuit from Pin to Pin

⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

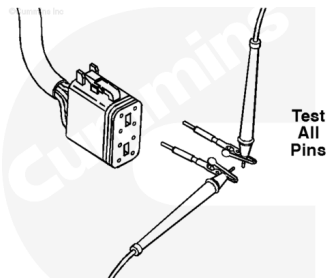


Turn the keyswitch to the OFF position. Disconnect the vehicle battery cables from the battery.

Note : Disconnecting the vehicle connections can require that the engine ECM real-time clock be reset using INSITE.

⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.



Disconnect the ICON™ idle control module A connector.

Touch one of the multimeter leads to pin 4 of the ICON™ idle control module A harness connector.

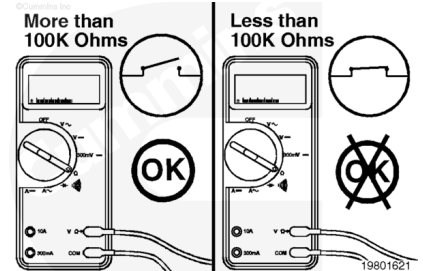
Touch the other multimeter lead to all other pins in the connector, one at a time.

Read the value displayed on the multimeter.

The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit from pin 4 to any other pin that registered a closed circuit.

Isolate the short circuit to the cab harness, cab thermostat jumper harness, or ICON™ engine harness.

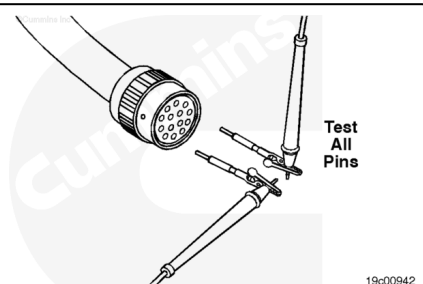


Disconnect the 14-pin pass-through connector at the vehicle's firewall.

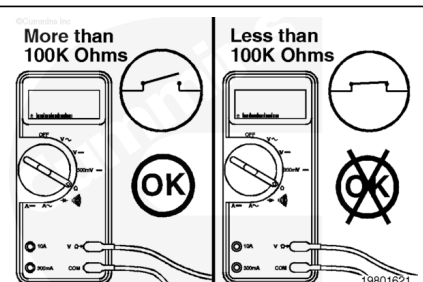
Touch one of the multimeter leads to pin B of the 14-pin pass-through connector, engine harness side.

Touch the other multimeter lead to all other pins in the connector, one at a time.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms, which is an open circuit. If the circuit is **not** open, there is a short circuit between pin B and any other pin that registered a closed circuit.



Repair or replace the ICON™ engine harness. Refer to Procedure 019-200 (/qs3/pubsys2/xml/en/procedures/97/97-019-200.html), 019-208 (/qs3/pubsys2/xml/en/procedures/97/97-019-208.html), or 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).

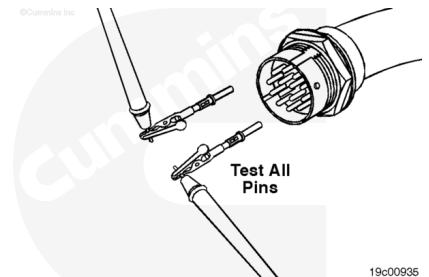
If the circuit is open, check the cab harness and cab thermostat jumper harness portion of the circuit.

Disconnect the cab thermostat jumper harness from the cab harness.

Touch one of the multimeter leads to pin B of the 14-pin pass-through connector, cab harness side.

Touch the other multimeter lead to all other pins in the connector, one at a time.

Read the value displayed on the multimeter.

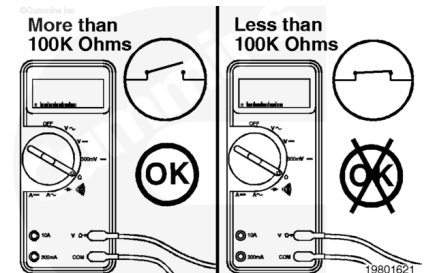


The multimeter **must** display a reading of more than 100k ohms, which is an open circuit.

If the circuit is **not** open, there is a short circuit between pin B and any other pin that registered a closed circuit.

Repair or replace the cab harness. Refer to Procedure 019-200 (/qs3/pubsys2/xml/en/procedures/97/97-019-200.html) or 019-305 (/qs3/pubsys2/xml/en/procedures/97/97-019-305.html).

If all pin-to-pin checks are open, isolate the short circuit to the cab thermostat jumper harness.

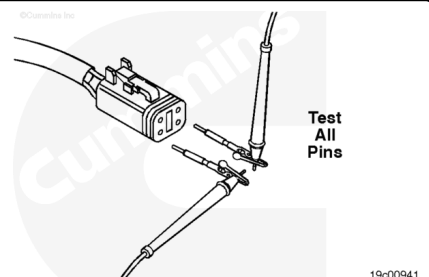


Disconnect the cab thermostat jumper harness connector from the cab harness.

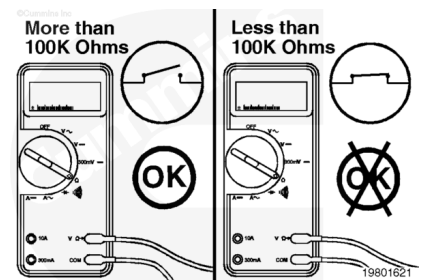
Touch one of the multimeter leads to pin 3 of the cab thermostat jumper harness connector, on the side that connects to the cab harness.

Touch the other multimeter lead to all other pins in the connector.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of more than 100k ohms. If the circuit is closed, there is a short circuit between pin 3 and any other pin that registered a closed circuit. Repair or replace the cab thermostat jumper harness. Refer to Procedure 019-204 (/qs3/pubsys2/xml/en/procedures/97/97-019-204.html) or 019-295 (/qs3/pubsys2/xml/en/procedures/97/97-019-295.html).



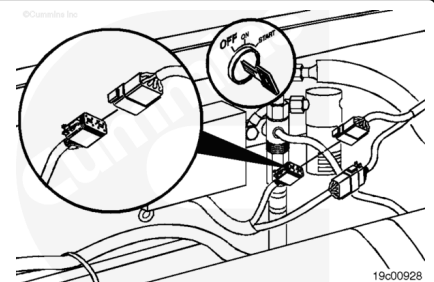
After repairing the cab harness connection end of the cab thermostat jumper harness, repeat the pin-to-pin check, pin 3 to all other pins. If the circuit is still closed, the thermostat end of the cab thermostat harness is the problem connector. Repair the connector or replace the cab thermostat jumper harness.

Connect all components after completing the repair.

Check for Short Circuit to External Voltage Source

⚠ CAUTION ⚠

To avoid pin and harness damage, use test leads, Part Number 3822758 and 3822917, when taking a measurement.



Turn the keyswitch to the OFF position.

Disconnect the ICON™ idle control module A connector.

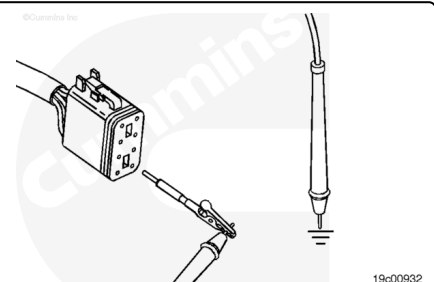
Set the multimeter to measure VDC.

Turn the keyswitch to the ON position.

Touch one of the multimeter leads to pin 4 of the ICON™ idle control module A harness connector.

Touch the other multimeter lead to ground.

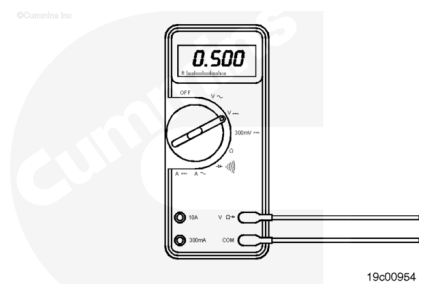
Read the value displayed on the multimeter.



The multimeter **must** display a reading of less than 0.5 VDC. If the voltage is **not** less than 0.5 VDC, there is a short circuit to an external voltage source.

Note : An external voltage source is any wire in the cab or ICON™ engine harness wiring that carries voltage.

Isolate the area of the short to external voltage source.

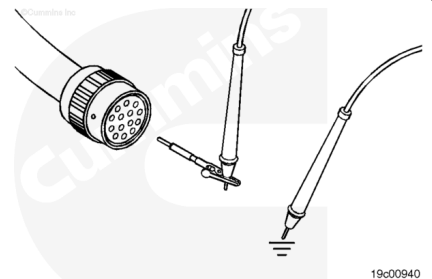


Disconnect the 14-pin pass-through connector at the vehicle's firewall.

Touch one of the multimeter leads to pin B of the 14-pin pass-through connector, engine harness side.

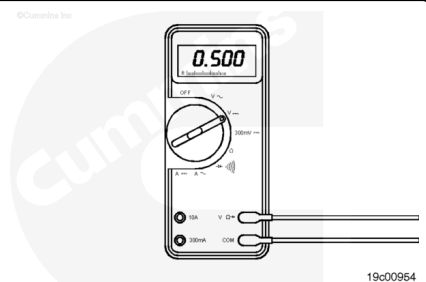
Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of less than 0.5 VDC. If the voltage is **not** less than 0.5 VDC, there is a short circuit to an external voltage source on the ICON™ engine harness side of the circuit. Remove the external voltage source.

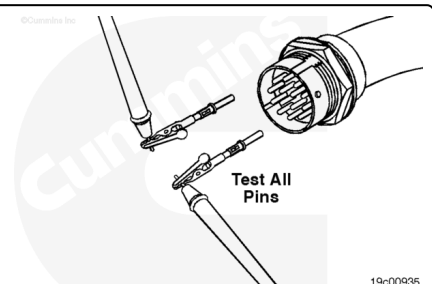
If the voltage is 0.5 VDC or less, the external voltage source is on the cab harness side of the circuit. Check the cab harness side of the circuit.



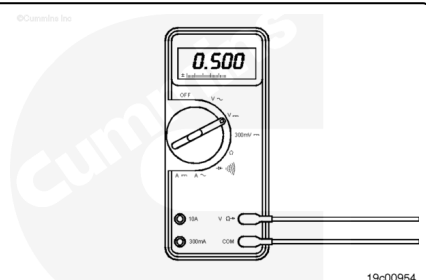
Touch one of the multimeter leads to pin B of the 14-pin pass-through connector, cab harness side.

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of less than 0.5 VDC. If the voltage is **not** less than 0.5 VDC, isolate the short circuit to external voltage source to the cab harness or the cab thermostat jumper harness.

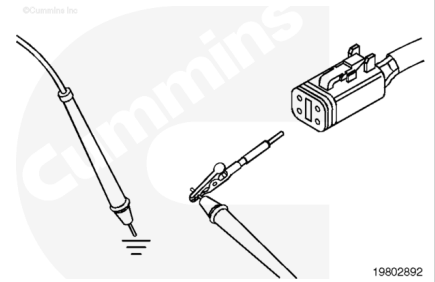


Disconnect the cab thermostat jumper harness from the cab harness.

Touch one of the multimeter leads to pin 3 of the cab thermostat jumper harness connector on the cab harness.

Touch the other multimeter lead to ground.

Read the value displayed on the multimeter.



The multimeter **must** display a reading of less than 0.5 VDC. If the voltage is **not** less than 0.5 VDC, there is a short circuit to an external voltage source in the cab harness.

If the voltage is 0.5 VDC or less, the short circuit to the external voltage source is originating from the cab thermostat jumper harness.

Remove the external voltage source.

Connect all components after completing the repair.

